Task1: Question 2 Two-way Table

Step 1: Define the variables clearly

We are studying **two categorical variables**:

1. **Website design**
   - Design 1
   - Design 2
2. **Whether the student watched the website for more than one minute**
   - Yes = watched for **more than 1 minute**
   - No = watched for **1 minute or less**

A table that summarizes the relationship between two categorical variables is called a **two-way table** (or $r \times c$**table**), where:

- r= number of rows
- c= number of columns

Here, there are:

- $r = 2$website designs
- $c = 2$watch-time categories

So this is a 2×2**table**.

Important assumption

The problem says that **50 students** were randomly assigned to one of the two website designs, and it gives the number in each design who watched for more than a minute:

- Design 1: 12 students watched more than a minute
- Design 2: 5 students watched more than a minute

To complete the full table, we need to know how many students were assigned to each design. Since the problem does not state separate group sizes, I will use the natural assumption that the 50 students were split evenly:

$$25 \text{ students assigned to Design 1, and } 25 \text{ students assigned to Design 2.}$$

(a) Translate the scenario into an $r \times c$table

Let:

- Row variable = **Website design**
- Column variable = **Watched for more than 1 minute?**

Since 25 students are assigned to each design:

- For **Design 1**, 12 watched more than a minute, so

$$25 - 12 = 13$$

  did **not** watch more than a minute.
- For **Design 2**, 5 watched more than a minute, so

$$25 - 5 = 20$$

  did **not** watch more than a minute.

Thus, the two-way table of **counts** is:

Website Design	$>$ 1 minute	$\leq$ 1 minute	Total
Design 1	12	13	25
Design 2	5	20	25
Total	17	33	50

Interpretation of the table

- 12 students assigned to Design 1 watched for more than 1 minute.
- 13 students assigned to Design 1 watched for 1 minute or less.
- 5 students assigned to Design 2 watched for more than 1 minute.
- 20 students assigned to Design 2 watched for 1 minute or less.

(b) Identify the joint distribution, the marginal distributions, and the conditional distributions

---

1. Joint distribution

The **joint distribution** gives the proportion of all 50 students that fall in each cell of the two-way table.

To find each joint proportion, divide the cell count by the total sample size 50.

Website Design	> 1 minute	$\leq$ 1 minute	Total
Design 1	$\frac{12}{50} = 0.24$	$\frac{13}{50} = 0.26$	$\frac{25}{50} = 0.50$
Design 2	$\frac{5}{50} = 0.10$	$\frac{20}{50} = 0.40$	$\frac{25}{50} = 0.50$
Total	$\frac{17}{50} = 0.34$	$\frac{33}{50} = 0.66$	1.00

So the **joint distribution** is:

- $P(\text{Design 1 and >1 minute}) = \dfrac{12}{50} = 0.24$
- $P(\text{Design 1 and ≤1 minute}) = \dfrac{13}{50} = 0.26$
- $P(\text{Design 2 and >1 minute}) = \dfrac{5}{50} = 0.10$
- $P(\text{Design 2 and ≤1 minute}) = \dfrac{20}{50} = 0.40$

In words

- 24% of all students were assigned to Design 1 **and** watched for more than a minute.
- 26% of all students were assigned to Design 1 **and** watched for one minute or less.
- 10% of all students were assigned to Design 2 **and** watched for more than a minute.
- 40% of all students were assigned to Design 2 **and** watched for one minute or less.

---

2. Marginal distributions

A **marginal distribution** is the distribution of one variable alone, ignoring the other variable. These are found from the row totals or column totals.

(i) Marginal distribution of website design

Using the row totals:

- $$P(\text{Design 1}) = \frac{25}{50} = 0.50$$
- $$P(\text{Design 2}) = \frac{25}{50} = 0.50$$

So the marginal distribution of website design is:

Design	Proportion
Design 1	0.50
Design 2	0.50

(ii) Marginal distribution of watch time

Using the column totals:

- More than 1 minute: $\frac{17}{50} = 0.34$
- 1 minute or less: $\frac{33}{50} = 0.66$

So the marginal distribution of watch time is:

Watch Time	Proportion
$>$ 1 minute	0.34
$\leq$ 1 minute	0.66

In words

- 50% of the students were assigned to each design.
- Overall, 34% of the students watched for more than a minute.
- Overall, 66% watched for one minute or less.

---

3. Conditional distributions

A **conditional distribution** describes the distribution of one variable **within a fixed category of the other variable**.

In this setting, the most useful conditional distributions are the distribution of **watch time given website design**, because website design is the explanatory variable and watch time is the response.

Conditional distribution of watch time given Design 1

Among the 25 students assigned to Design 1:

- Proportion who watched more than 1 minute:

$$P(>\text{1 minute} \mid \text{Design 1}) = \frac{12}{25} = 0.48$$

- Proportion who watched 1 minute or less:

$$P(\leq \text{1 minute} \mid \text{Design 1}) = \frac{13}{25} = 0.52$$

So for Design 1:

Watch Time	Conditional Proportion
$>$ 1 minute	0.48
$\leq$ 1 minute	0.52

Conditional distribution of watch time given Design 2

Among the 25 students assigned to Design 2:

- Proportion who watched more than 1 minute:

$$P(>1 \text{ minute} \mid \text{Design 2}) = \frac{5}{25} = 0.20$$

- Proportion who watched 1 minute or less:

$$P(\leq 1 \text{ minute} \mid \text{Design 2}) = \frac{20}{25} = 0.80$$

So for Design 2:

Watch Time	Conditional Proportion
$>$ 1 minute	0.20
$\leq$ 1 minute	0.80

Summary of the conditional distributions

Design	$>$ 1 minute	$\leq$ 1 minute
Design 1	0.48	0.52
Design 2	0.20	0.80

Interpretation

- For **Design 1**, 48% watched for more than one minute.
- For **Design 2**, only 20% watched for more than one minute.

This suggests that **Design 1 appears to hold attention better** than Design 2.

Optional: Reverse conditional distributions

Among those who watched more than 1 minute (17 students total):

- $$P(\text{Design 1} \mid >1 \text{ minute}) = \frac{12}{17} \approx 0.706$$
- $$P(\text{Design 2} \mid >1 \text{ minute}) = \frac{5}{17} \approx 0.294$$

Among those who watched 1 minute or less (33 students total):

- $$P(\text{Design 1} \mid \text{<1 minute}) = \frac{13}{33} \approx 0.394$$
- $$P(\text{Design 2} \mid \text{<1 minute}) = \frac{20}{33} \approx 0.606$$

These are correct as well, but the more standard conditional distributions for comparing designs are the ones computed **within each design**.

Final answer

(a) $r \times c$ table

Let the rows represent the two website designs and the columns represent whether a student watched the website for more than one minute.

Assuming the 50 students were split evenly between the two designs, there were 25 students in each group.

Website Design	> 1 minute	$\leq$ 1 minute	Total
Design 1	12	13	25
Design 2	5	20	25
Total	17	33	50

(b) Joint, marginal, and conditional distributions

Joint distribution

The joint distribution is the set of cell proportions out of 50:

Website Design	> 1 minute	$\leq$ 1 minute
Design 1	$\frac{12}{50} = 0.24$	$\frac{13}{50} = 0.26$
Design 2	$\frac{5}{50} = 0.10$	$\frac{20}{50} = 0.40$

Marginal distributions

- Website design:

$$P(\text{Design 1}) = \frac{25}{50} = 0.50, P(\text{Design 2}) = \frac{25}{50} = 0.50$$

- Watch time:

$$P(\text{>1 minute}) = \frac{17}{50} = 0.34, P(\leq \text{1 minute}) = \frac{33}{50} = 0.66$$

Conditional distributions of watch time given design

For Design 1:

$$P(>1 \text{ minute} \mid \text{Design 1}) = \frac{12}{25} = 0.48, P(\leq 1 \text{ minute} \mid \text{Design 1}) = \frac{13}{25} = 0.52$$

For Design 2:

$$P(>1 \text{ minute} \mid \text{Design 2}) = \frac{5}{25} = 0.20, P(\leq 1 \text{ minute} \mid \text{Design 2}) = \frac{20}{25} = 0.80$$

Therefore, the conditional distributions show that students assigned to **Design 1** were more likely to watch the site for more than one minute than students assigned to **Design 2**.

Task 2: Question 5

We are given two possible discrete probability distributions for a random variable X, called $p_0$and p_1.

The probability distributions are:

x	0	1	2	3	4	5	6
$p_0(x)$	0.1	0.1	0.2	0.3	0.1	0.1	0.1
$p_1(x)$	0.1	0.3	0.2	0.1	0.1	0.1	0.1

We want to test

$$H_0 : p_0 \text{ is the correct distribution}$$

against

$$H_a : p_1 \text{ is the correct distribution}$$

The decision rule is:

Reject H_0only if $X \leq 1$.

That means the **rejection region** is

$$R = \{0,1\}.$$

Step 1: Define the important terms

Null hypothesis

The **null hypothesis**, denoted by H_0, is the statement we begin by assuming is true unless the data provide evidence against it.

Here:

$$H_0 : X \text{ has distribution } p_0.$$

Alternative hypothesis

The **alternative hypothesis**, denoted by H_a, is the competing claim.

Here:

$$H_a: X \text{ has distribution } p_1 .$$

Type I error

A **Type I error** occurs when we **reject H_0 even though H_0 is actually true**.

So here, a Type I error means:

- the true distribution is p_0,
- but we reject H_0.

Type II error

A **Type II error** occurs when we **fail to reject H_0 even though H_a is true**.

So here, a Type II error means:

- the true distribution is p_1,
- but we do **not** reject H_0.

Step 2: Understand the rejection rule

We reject H_0 only if

$$X \leq 1.$$

Since X can take the values $0, 1, 2, 3, 4, 5, 6$, the rejection region is:

$$\{0, 1\} .$$

So:

- If $X = 0$ or $X = 1$, we reject H_0.
- If $X = 2, 3, 4, 5,$ or 6, we do **not** reject H_0.

Part (a): Find the probability of a Type I error

Definition

The probability of a Type I error is

$$P(\text{reject } H_0 \mid H_0 \text{ is true}).$$

Because H_0 says the correct distribution is p_0, we calculate this probability using the probabilities in the p_0 row.

Since we reject H_0 when $X \leq 1$,

$$P(\text{Type I error}) = P(X \leq 1 \mid p_0).$$

Using the table for p_0:

$$p_0(0) = 0.1, p_0(1) = 0.1.$$

Therefore,

$$\begin{aligned} P(X \leq 1 \mid p_0) &= P(X = 0 \mid p_0) + P(X = 1 \mid p_0) \\ &= 0.1 + 0.1 = 0.2. \end{aligned}$$

Answer to part (a)

$$P(\text{Type I error}) = 0.20$$

Interpretation

If p_0 is actually the correct distribution, then this testing rule will incorrectly reject H_0 with probability **0.20**, or **20%**.

Part (b): Find the probability of a Type II error

Definition

The probability of a Type II error is

$$P(\text{do not reject } H_0 \mid H_a \text{ is true}).$$

Because H_a says the correct distribution is p_1, we calculate this probability using the probabilities in the p_1 row.

A Type II error happens when p_1 is true, but we **fail to reject** H_0. Since we reject only when $X \leq 1$, we fail to reject when

$$X > 1.$$

Thus,

$$P(\text{Type II error}) = P(X > 1 \mid p_1).$$

The values greater than 1 are:

$$2, 3, 4, 5, 6.$$

From the p_1 row:

$$p_1(2) = 0.2, p_1(3) = 0.1, p_1(4) = 0.1, p_1(5) = 0.1, p_1(6) = 0.1\,.$$

So,

$$P(X > 1 \mid p_1) = 0.2 + 0.1 + 0.1 + 0.1 + 0.1 = 0.6\,.$$

Answer to part (b)

$$P(\text{Type II error}) = 0.60$$

Interpretation

If p_1 is actually the correct distribution, then this rule fails to reject H_0 with probability **0.60**, or **60%**.

Final Answers

$$\text{(a) Type I error probability } = 0.20$$
$$\text{(b) Type II error probability } = 0.60$$

Short conclusion

The rejection region is $R = \{0,1\}$, since the rule says to reject H_0 only when $X \leq 1$.

- The probability of a **Type I error** is the probability of rejecting H_0 when p_0 is true:

  $$P(X \leq 1 \mid p_0) = p_0(0) + p_0(1) = 0.1 + 0.1 = 0.2\,.$$

- The probability of a **Type II error** is the probability of not rejecting H_0 when p_1 is true:

  $$P(X > 1 \mid p_1) = p_1(2) + p_1(3) + p_1(4) + p_1(5) + p_1(6) = 0.2 + 0.1 + 0.1 + 0.1 + 0.1 = 0.6\,.$$

Therefore,

$$\alpha = 0.20 \text{ and } \beta = 0.60\,.$$

- α is the **Type I error probability**
- β is the **Type II error probability**
- The **power** of the test is

  $$1 - \beta = 1 - 0.6 = 0.4$$

  so the test correctly rejects H_0 when p_1 is true with probability **0.40**.

Task 3: Is there a random distribution of trees? (IPS Exercise 9.26)

We are told that a random sample of **100 trees** was taken from the Wade Tract, and the tract was divided into **four equal parts** (labeled Q_1, Q_2, Q_3, Q_4). The numbers of sampled trees in each part were:

Quadrant	Q_1	Q_2	Q_3	Q_4
Count	18	22	39	21

We want to assess whether these counts are consistent with a **random distribution** of trees.

(a) If the trees are randomly distributed, why do we expect 25 trees in each quadrant?

Key idea

If the trees are **randomly distributed** over the tract, and the tract is divided into **four equal-area parts**, then each tree is equally likely to fall into any one of the four quadrants.

Let

$$p_i = P(\text{a randomly chosen tree falls in quadrant } Q_i), i = 1,2,3,4\,.$$

Because the four parts are equal in area and the distribution is assumed random,

$$p_1 = p_2 = p_3 = p_4 = \frac{1}{4}\,.$$

Now let $n = 100$be the sample size. The **expected count** in each quadrant is

$$E_i = np_i = 100\left(\frac{1}{4}\right) = 25.$$

So, if the trees are randomly distributed, we expect **25 trees in each quadrant on average**.

Answer to part (a)

If the trees are randomly distributed and the four quadrants have equal area, then each tree has probability $1/4$of landing in any quadrant. Therefore, in a sample of 100 trees, the expected number in each quadrant is

$$100 \times \frac{1}{4} = 25.$$

(b) Why do we not expect to get exactly 25 trees in each quadrant?

Even if the trees truly are randomly distributed, the observed counts from a **sample** will usually vary from the expected counts because of **chance variation** (also called **sampling variability**).

The expected count of 25 in each quadrant is a **long-run average**, not a guarantee for every sample. In real data, random fluctuations cause some quadrants to have slightly more than 25 and others slightly fewer than 25.

So, a random distribution does **not** mean we must observe exactly $25, 25, 25, 25$. Instead, it means the observed counts should be **reasonably close** to those expected under randomness.

Answer to part (b)

We do not expect exactly 25 trees in each quadrant because the sample counts are subject to random variation. The value 25 is the **expected count**, not a count that must occur exactly in every sample. Small differences from 25 are normal when sampling randomly.

(c) Perform the goodness-of-fit test

Now we formally test whether the observed counts are consistent with a random distribution.

Step 1: State the hypotheses

This is a **chi-square goodness-of-fit test**.

Null hypothesis

The **null hypothesis** says that the trees are randomly distributed among the four equal quadrants, so each quadrant has probability $1/4$:

$$H_0: p_1 = p_2 = p_3 = p_4 = 0.25$$

where p_i is the probability that a randomly chosen tree falls in quadrant Q_i.

Alternative hypothesis

The **alternative hypothesis** says that the distribution is **not** random in this sense; that is, at least one quadrant has a different probability:

$$H_a: \text{Not all } p_i \text{ are equal to } 0.25$$

Step 2: Check conditions for the chi-square goodness-of-fit test

For a chi-square goodness-of-fit test, we need the **expected counts** to be large enough. A standard guideline is that all expected counts should be at least 5.

Here, the expected counts are:

$$E_1 = E_2 = E_3 = E_4 = 25$$

Since all expected counts are greater than 5, the condition is satisfied.

Step 3: Compute the test statistic

The chi-square goodness-of-fit test statistic is

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

- O_i= observed count in category i
- E_i= expected count in category i

Observed and expected counts

Quadrant	Q_1	Q_2	Q_3	Q_4
O_i	18	22	39	21
E_i	25	25	25	25

Now compute each contribution:

For Q_1

$$\frac{(18-25)^2}{25} = \frac{(-7)^2}{25} = \frac{49}{25} = 1.96$$

For Q_2

$$\frac{(22-25)^2}{25} = \frac{(-3)^2}{25} = \frac{9}{25} = 0.36$$

For Q_3

$$\frac{(39-25)^2}{25} = \frac{14^2}{25} = \frac{196}{25} = 7.84$$

For Q_4

$$\frac{(21-25)^2}{25} = \frac{(-4)^2}{25} = \frac{16}{25} = 0.64$$

Total chi-square statistic

$$\chi^2 = 1.96 + 0.36 + 7.84 + 0.64 = 10.80$$

So the test statistic is

$$\chi^2 = 10.80$$

Step 4: Degrees of freedom

For a chi-square goodness-of-fit test with $k$categories,

$$df = k - 1$$

provided that no parameters are estimated from the data.

Here there are $k = 4$quadrants, so

$$df = 4 - 1 = 3$$

Thus,

$$df = 3$$

Step 5: Find the p-value

The p-value is the probability, assuming the null hypothesis is true, of obtaining a chi-square statistic at least as large as the observed value.

Using $\chi^2 = 10.80$with $df = 3$,

$$p\text{-value} \approx 0.0129$$

So,

$$p \approx 0.013$$

Step 6: State the decision

Using the common significance level $\alpha = 0.05$,

$$p = 0.013 < 0.05$$

Therefore, we **reject** H_0.

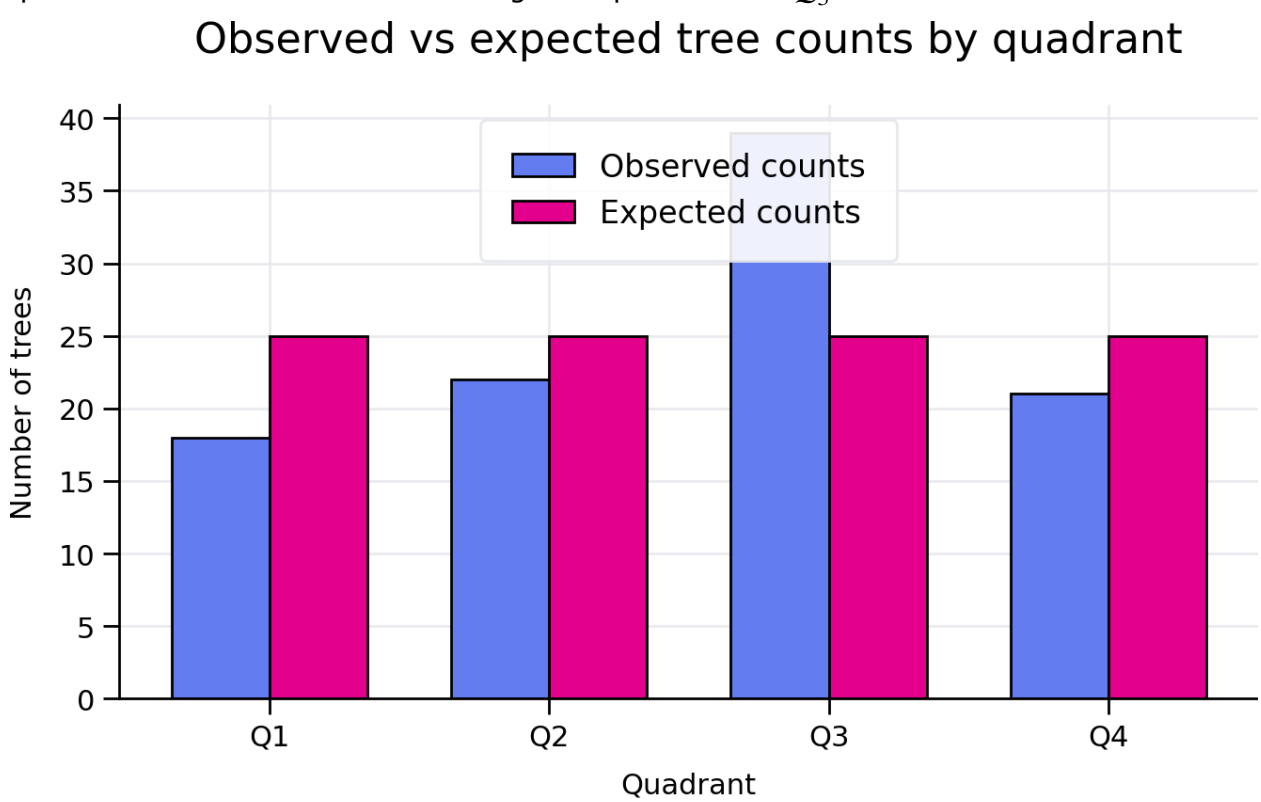
Step 7: Conclusion in context

There is statistically significant evidence that the trees are **not** distributed equally among the four quadrants. Therefore, the data are **not consistent with a random scattering of trees** across these equal parts of the tract.

In particular, quadrant Q_3has **39 trees**, which is much larger than the expected count of 25, and this quadrant contributes the most to the chi-square statistic. This suggests that the trees may be **clustered** rather than randomly scattered.

Graph: observed counts versus expected counts

The graph below helps visualize the result. If the trees were randomly distributed, each quadrant would be near 25. The largest departure is in Q_3.



Short report

A chi-square goodness-of-fit test was used to determine whether the 100 sampled trees were randomly distributed among the four equal quadrants.

Let $p_i$denote the probability that a randomly selected tree falls in quadrant Q_i. If the trees are randomly distributed, then because the quadrants are equal in area,

$$H_0: p_1 = p_2 = p_3 = p_4 = 0.25\,.$$

The alternative hypothesis is that the probabilities are not all equal to 0.25.

The observed counts were

$$(18,22,39,21),$$

and the expected counts under H_0were

$$(25,25,25,25)\,.$$

All expected counts are at least 5, so the chi-square test is appropriate.

The test statistic is

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \frac{(18-25)^2}{25} + \frac{(22-25)^2}{25} + \frac{(39-25)^2}{25} + \frac{(21-25)^2}{25} = 10.80\,.$$

The degrees of freedom are

$$df = 4 - 1 = 3.$$

The p-value is approximately

$$p = 0.013\,.$$

Because $p < 0.05$, we reject H_0. There is significant evidence that the trees are not randomly scattered among the four quadrants. The unusually large count in Q_3suggests clustering.

Final answers

(a)

If the trees are randomly distributed and the tract is divided into four equal-area quadrants, each tree has probability $1/4$of falling in any quadrant. Therefore, in a sample of 100 trees, the expected count in each quadrant is $100(1/4) = 25$.

(b)

We do not expect exactly 25 trees in each quadrant because of random sampling variability. The value 25 is the expected count, not a guaranteed count in every sample.

(c)

Use a chi-square goodness-of-fit test.

$$H_0: p_1 = p_2 = p_3 = p_4 = 0.25, H_a: \text{not all } p_i = 0.25$$

Observed counts: $(18, 22, 39, 21)$

Expected counts: $(25, 25, 25, 25)$

$$\chi^2 = \frac{(18-25)^2}{25} + \frac{(22-25)^2}{25} + \frac{(39-25)^2}{25} + \frac{(21-25)^2}{25} = 10.80$$
$$df = 4 - 1 = 3$$
$$p \approx 0.013$$

Since $p < 0.05$, reject H_0. There is significant evidence that the trees are not randomly distributed among the four quadrants.

Task 4: Joint probability table for two wine tasters

Two tasters each rate a randomly chosen wine on a scale from 1 to 5.

Let:

- X= rating given by **Taster 1**
- Y= rating given by **Taster 2**

The table gives the **joint probability distribution** of (X, Y). That is, each entry in the table is a probability of the form

$$P(X = x,\ Y = y)$$

for ratings $x, y \in \{1,2,3,4,5\}$.

The table is:

	\multicolumn$5c$Taster 2				
Taster 1	1	2	3	4	5
1	0.03	0.02	0.01	0.00	0.00
2	0.02	0.07	0.06	0.02	0.01
3	0.01	0.05	0.25	0.05	0.01
4	0.00	0.02	0.05	0.20	0.02
5	0.00	0.01	0.01	0.02	0.06

(a) Why is this a legitimate assignment of probabilities to outcomes?

A legitimate probability distribution must satisfy **two conditions**:

Condition 1: Every probability must be between 0 and 1

Each table entry is nonnegative and lies between 0 and 1:

- for example, $0.03, 0.07, 0.25, 0.20, 0.06$, etc.
- there are no negative probabilities
- there are no probabilities greater than 1

So the first condition is satisfied.

Condition 2: The probabilities of all possible outcomes must add to 1

Now add all entries in the table.

Row 1 total

$$0.03 + 0.02 + 0.01 + 0.00 + 0.00 = 0.06$$

Row 2 total

$$0.02 + 0.07 + 0.06 + 0.02 + 0.01 = 0.18$$

Row 3 total

$$0.01 + 0.05 + 0.25 + 0.05 + 0.01 = 0.37$$

Row 4 total

$$0.00 + 0.02 + 0.05 + 0.20 + 0.02 = 0.29$$

Row 5 total

$$0.00 + 0.01 + 0.01 + 0.02 + 0.06 = 0.10$$

Now sum the row totals:

$$0.06 + 0.18 + 0.37 + 0.29 + 0.10 = 1.00$$

So the total probability is 1.

Answer to part (a)

Yes, this is a legitimate assignment of probabilities because:

1. every probability in the table is between 0 and 1, and
2. the probabilities of all possible rating pairs sum to 1.

Therefore, the table is a valid **joint probability distribution** for (X, Y).

(b) What is the probability that the tasters agree when tasting wine?

The tasters **agree** when they give the same rating. That means:

$$X = Y$$

These are the diagonal entries of the table:

- $P(X = 1, Y = 1) = 0.03$
- $P(X = 2, Y = 2) = 0.07$
- $P(X = 3, Y = 3) = 0.25$
- $P(X = 4, Y = 4) = 0.20$
- $P(X = 5, Y = 5) = 0.06$

So,

$$\begin{aligned} P(X = Y) &= 0.03 + 0.07 + 0.25 + 0.20 + 0.06 \\ &= 0.61 \end{aligned}$$

Answer to part (b)

$$P(\text{tasters agree}) = 0.61$$

Interpretation

The two tasters give the same rating **61% of the time**.

(c) What is the probability that Taster 1 rates a wine higher than 3? What is the probability that Taster 2 rates a wine higher than 3?

"Higher than 3" means a rating of **4 or 5**.

Probability that Taster 1 rates the wine higher than 3

We want:

$$P(X > 3) = P(X = 4) + P(X = 5)$$

To find $P(X = 4)$, sum across the row for $X = 4$:

$$P(X = 4) = 0.00 + 0.02 + 0.05 + 0.20 + 0.02 = 0.29$$

To find $P(X = 5)$, sum across the row for $X = 5$:

$$P(X = 5) = 0.00 + 0.01 + 0.01 + 0.02 + 0.06 = 0.10$$

Therefore,

$$P(X > 3) = 0.29 + 0.10 = 0.39$$

So,

$$P(\text{Taster 1 rates higher than 3}) = 0.39$$

Probability that Taster 2 rates the wine higher than 3

We want:

$$P(Y > 3) = P(Y = 4) + P(Y = 5)$$

To find $P(Y = 4)$, sum down the column for $Y = 4$:

$$P(Y = 4) = 0.00 + 0.02 + 0.05 + 0.20 + 0.02 = 0.29$$

To find $P(Y = 5)$, sum down the column for $Y = 5$:

$$P(Y = 5) = 0.00 + 0.01 + 0.01 + 0.02 + 0.06 = 0.10$$

Therefore,

$$P(Y > 3) = 0.29 + 0.10 = 0.39$$

So,

$$P(\text{Taster 2 rates higher than 3}) = 0.39$$

Answer to part (c)

$$P(X > 3) = 0.39 \text{and} P(Y > 3) = 0.39$$

Interpretation

Each taster gives a rating above 3 with probability **0.39**.

(d) Taster 1's rating for a wine is 3. What is the conditional probability that Taster 2's rating is higher than 3?

This is a **conditional probability** question.

We want:

$$P(Y > 3 \mid X = 3)$$

That means:

Among the wines that Taster 1 rated as 3, what proportion did Taster 2 rate above 3?

Step 1: Find $P(X = 3)$

This is the total of the row where $X = 3$:

$$P(X = 3) = 0.01 + 0.05 + 0.25 + 0.05 + 0.01 = 0.37$$

Step 2: Find $P(Y > 3 \text{ and } X = 3)$

"$Y > 3$" means $Y = 4$or $Y = 5$. In the row $X = 3$, those entries are:

- $$P(X = 3, Y = 4) = 0.05$$
- $$P(X = 3, Y = 5) = 0.01$$

So,

$$P(Y > 3 \text{ and } X = 3) = 0.05 + 0.01 = 0.06$$

Step 3: Apply the conditional probability formula

The conditional probability formula is

$$P(Y > 3 \mid X = 3) = \frac{P(Y > 3 \text{ and } X = 3)}{P(X = 3)}$$

Substitute the values:

$$P(Y > 3 \mid X = 3) = \frac{0.06}{0.37}$$
$$\approx 0.1622$$

So,

$$P(Y > 3 \mid X = 3) = \frac{0.06}{0.37} \approx 0.162$$

Answer to part (d)

$$P(\text{Taster 2 rates higher than 3} \mid \text{Taster 1 rates } 3) \approx 0.162$$

Interpretation

Given that Taster 1 rated the wine as 3, the probability that Taster 2 rated it above 3 is about **0.162**, or **16.2%**.

(e) What is the probability that Taster 1's rating is higher than Taster 2's rating?

We want:

$$P(X > Y)$$

That means we add all entries where the row number is greater than the column number; in other words, all cells **below the main diagonal**.

These are:

- For $X = 2$: $Y = 1$

$$P(2,1) = 0.02$$

- For $X = 3$: $Y = 1,2$

$$P(3,1) = 0.01, P(3,2) = 0.05$$

- For $X = 4$: $Y = 1,2,3$

$$P(4,1) = 0.00, P(4,2) = 0.02, P(4,3) = 0.05$$

- For $X = 5$: $Y = 1,2,3,4$

$$P(5,1) = 0.00, P(5,2) = 0.01, P(5,3) = 0.01, P(5,4) = 0.02$$

Now add them:

$$P(X > Y) = 0.02 + 0.01 + 0.05 + 0.00 + 0.02 + 0.05 + 0.00 + 0.01 + 0.01 + 0.02 \\ = 0.19$$

So,

$$P(X > Y) = 0.19$$

Answer to part (e)

$$P(\text{Taster 1 rates higher than Taster 2}) = 0.19$$

Interpretation

Taster 1 gives a higher rating than Taster 2 with probability **0.19**, or **19%**.

Final answers

(a)

This is a legitimate probability distribution because every entry is between 0 and 1, and all entries sum to 1.

$$0.06 + 0.18 + 0.37 + 0.29 + 0.10 = 1.00$$

(b)

The tasters agree when $X = Y$, so add the diagonal entries:

$$0.03 + 0.07 + 0.25 + 0.20 + 0.06 = 0.61$$
$$0.61$$

(c)

Taster 1 rates higher than 3:

$$P(X > 3) = P(X = 4) + P(X = 5) = 0.29 + 0.10 = 0.39$$

Taster 2 rates higher than 3:

$$P(Y > 3) = P(Y = 4) + P(Y = 5) = 0.29 + 0.10 = 0.39$$
$$0.39 \text{ and } 0.39$$

(d)

$$P(Y > 3 \mid X = 3) = \frac{P(X = 3, Y = 4) + P(X = 3, Y = 5)}{P(X = 3)}$$
$$= \frac{0.05 + 0.01}{0.37} = \frac{0.06}{0.37} \approx 0.162$$
$$0.162$$

(e)

Add all probabilities where Taster 1's rating exceeds Taster 2's:

$$0.02 + 0.01 + 0.05 + 0.00 + 0.02 + 0.05 + 0.00 + 0.01 + 0.01 + 0.02 = 0.19$$
$$0.19$$

Short conclusion

Let X be Taster 1's rating and Y be Taster 2's rating. The given table is a valid joint distribution because all probabilities are between 0 and 1 and they sum to 1. The probability that the tasters agree is the sum of the diagonal probabilities, which is 0.61. The probability that either taster gives a rating above 3 is 0.39. The conditional probability that Taster 2 rates above 3 given that Taster 1 rates 3 is

$$\frac{0.06}{0.37} \approx 0.162\,.$$

Finally, the probability that Taster 1 gives a higher rating than Taster 2 is 0.19.